

(allow 1 mark for 'bullet embeds itself in block' etc.)

4	(a) brittle	B1 [1]
	(b) (i) stress = force/area = $60/(7.9 \times 10^{-7})$ = 7.6×10^7 Pa	C1 A1 [2]
	(ii) Young modulus = stress/strain limiting strain = $0.03/24$ ($= 1.25 \times 10^{-3}$) Young modulus = $(7.6 \times 10^7)/(1.25 \times 10^{-3}) = 6.1 \times 10^{10}$ Pa	C1 C1 A1 [3]
	(iii) energy = $\frac{1}{2} \times 60 \times 3.0 \times 10^{-4}$ = 9.0×10^{-3} J	C1 A1 [2]
	(c) If hard, ball does not deform (much) and <u>either</u> (all) kinetic energy converted to strain energy If soft, E_k becomes strain energy of ball and window (no mention of <u>strain</u> energy, max 2 marks) <u>or</u> impulse for hard ball takes place over shorter time (B1) larger force/greater stress (B1)	B1 B1 B1 [3]

Page 2	Mark Scheme	Syllabus	Paper
	A and AS LEVEL – June 2005	9702	2